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AquaShield-AI: A Closed-Loop IoT Framework Integrating Biodegradable Shade and Soil-Additive Treatments for Precision Irrigation Water Conservation

Abstract*—*Water scarcity increasingly constrains smallholder and school-scale agriculture, motivating low-cost interventions that reduce irrigation demand without depending on non-biodegradable materials. This paper presents AquaShield-AI, an Internet of Things (IoT) framework that combines biodegradable shading and fruit-peel-derived soil-additive treatments within a closed-loop, ESP32-based irrigation controller. A 3×3 factorial field design (shade × additive, n = 3 replicates) is coupled with a two-layer control law that unites moisture-threshold actuation with a rain-aware predictive layer, MQTT telemetry, and LoRaWAN connectivity for farm-scale deployment. A 30-day water-balance simulation, parameterized from literature-reported treatment-effect magnitudes, is analyzed with two-way ANOVA and Random Forest feature importance. Results show statistically significant main effects of shade ($p < 0.001$) and additive ($p < 0.001$) on irrigation water use, no significant interaction ($p = 0.588$), and up to 58.3% simulated water savings, with the fully biodegradable combination recovering 77.9% of the non-biodegradable benchmark's savings.

Index Terms*—*closed-loop irrigation control, IoT, precision agriculture, biodegradable soil additive, ESP32, factorial design, MQTT

I. INTRODUCTION

Agriculture accounts for most global freshwater withdrawals, and rising demand under increasingly variable rainfall has intensified interest in low-cost, field-deployable methods for reducing irrigation water use, particularly for smallholder and school-scale producers who cannot absorb the capital cost of conventional precision-irrigation infrastructure [1].

Two complementary interventions have been studied for this purpose. The first is surface shading or mulching, which reduces evaporative demand at the soil surface; the second is the incorporation of water-retaining soil additives, which increases the effective water-holding capacity of the root zone. Internet of Things (IoT) platforms built around low-cost microcontrollers have been used extensively to automate the measurement side of this problem: ESP32- and ESP8266-based systems have demonstrated soil-moisture sensing, cloud telemetry, and threshold-triggered pump control at low hardware cost [1], [2]. However, most reported systems evaluate a single intervention either a mulch material or an additive in isolation and actuate irrigation using a single reactive threshold rule without accounting for imminent rainfall or bounding the failure modes of an automated valve.

This creates two open problems that motivate the present work. First, from an agronomic standpoint, it is not established whether combining a biodegradable shading treatment with a biodegradable water-retaining additive produces an additive, synergistic, or redundant reduction in irrigation demand a question a single-factor trial cannot answer. Second, from a systems-engineering standpoint, a controller that acts on its own sensor data (opening and closing a physical valve) introduces failure modes a stuck-open valve, a network outage, a runaway control decision that a passive monitoring-only IoT deployment does not have to address, yet these fail-safe requirements are rarely reported alongside the control logic itself.

This paper addresses both problems with AquaShield-AI, an IoT framework that (i) evaluates biodegradable shade and soil-additive treatments in a single 3×3 factorial design rather than as separate single-factor studies; (ii) implements a two-layer closed-loop irrigation controller on a commodity ESP32 module, combining a moisture-threshold baseline with a rain-aware predictive layer; (iii) enforces firmware-level fail-safes a maximum continuous actuation time, a daily watering cap, and a watchdog timer so that a network or logic fault cannot leave the valve open indefinitely; and (iv) validates the resulting system through a parameterized water-balance simulation analyzed with two-way ANOVA and Random Forest feature importance, rather than asserting performance without evidence.

The remainder of this paper is organized as follows. Section II reviews related work on IoT-based irrigation and biodegradable water-conservation materials and identifies the specific research gap this paper addresses. Section III describes the six-layer system architecture, the water-balance and control-law mathematical models, and the algorithmic workflow. Section IV details the experimental setup and simulation methodology, including hardware, tools, and parameters. Section V presents the simulation results, including the factorial analysis, water-savings comparison, and rain-classifier performance. Section VI concludes the paper and outlines its limitations and future work.

II. BACKGROUND AND MOTIVATION

A. IoT-Based Irrigation Automation

Low-cost microcontroller platforms have made automated, sensor-driven irrigation universally accessible. Vaishali et al. demonstrated a mobile-integrated smart irrigation and monitoring system that combines soil-moisture sensing with cloud reporting for remote decision-making [2], and subsequent ESP32-based designs have added waterproof soil-temperature probes, air temperature/humidity sensing, and flow measurement to the same basic sense-and-report architecture, most reaching a decision rule no more sophisticated than a single moisture threshold that opens a pump or valve [1]. This body of work establishes that the hardware and connectivity layer for low-cost automated irrigation is mature; what it does not establish is how such a system should behave

when its own actuation can fail, or when its irrigation decision should be withheld because rain is already supplying water both of which are addressed in Section III of this paper.

B. Biodegradable Shading and Soil-Additive Materials

Independently of the IoT literature, materials-science and agronomy studies have evaluated biodegradable alternatives to plastic mulch and synthetic superabsorbent polymers. El-Beltagi et al. review the water-conservation benefits of organic and biodegradable mulching materials, reporting that mulching reduces surface evaporation, moderates soil temperature, and measurably increases water-use efficiency relative to bare soil, while noting that plastic mulch though still the most common commercial choice leaves persistent field residues that biodegradable alternatives avoid [3]. On the additive side, Fang et al. synthesized superabsorbent hydrogels from melon- and orange-peel waste and reported swelling capacities up to 765.6 g/g and a sandy-soil water-retention increase of up to 271% relative to untreated soil, outperforming an unmodified control [4]. These results establish that both interventions are independently effective, but neither study nor the broader literature they represent evaluates the two treatments together in a single factorial trial, or couples' treatment to an automated irrigation decision.

C. Connectivity for Farm-Scale Deployment

A system validated on a single instrumented plot does not automatically generalize to a multi-acre deployment. Singh et al. deployed a real-time, end-to-end LoRaWAN sensor network across a commercial greenhouse and reported successful long-range, low-power operation as an alternative to Wi-Fi-dependent designs, while explicitly noting that wireless sensor network coverage and enclosure design remain practical bottlenecks for precision-agriculture deployment at scale [5]. This motivates LoRaWAN as a connectivity option in the present architecture (Section III-D) rather than assuming Wi-Fi availability.

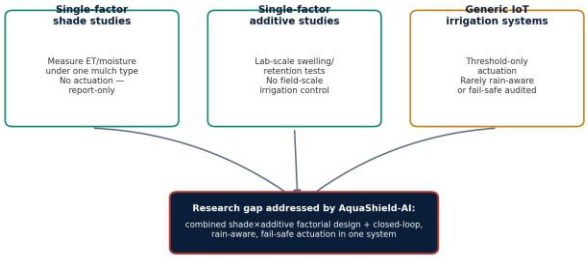


Fig. 1. Research gap addressed by AquaShield-AI relative to single-factor shade studies, single-factor additive studies, and generic threshold-only IoT irrigation systems.

D. Identified Research Gap

Synthesizing the three strands above (Fig. 1), the specific gap this paper addresses is the absence of a system that (i) evaluates biodegradable shade and additive treatments jointly, so that any interaction effect between them can be tested statistically rather than assumed; and (ii) actuates irrigation from that combined evidence through a controller whose failure modes are explicitly bounded, rather than through an unaudited single threshold rule. Section III presents the architecture designed to close this gap.

III. SYSTEM ARCHITECTURE

AquaShield-AI is organized into six layers spanning material preparation, field sensing, edge actuation, connectivity, cloud analytics, and statistical decision support, shown in Fig. 2. Layers 1–3 are physically deployed per plot; Layers 4–6 are shared across the full 3×3 factorial deployment.

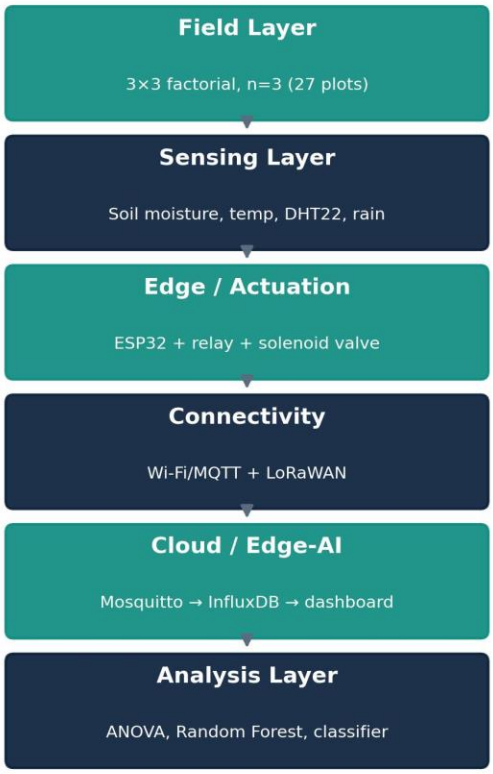


Fig. 2. AquaShield-AI six-layer system architecture, from field-layer sensing through cloud analytics and statistical decision support.

A. Soil-Moisture Water-Balance Model

Each plot's root-zone volumetric water content is modeled as a discrete-time water balance updated once per day (or per control cycle in firmware),

$$\theta(t+1) = \theta(t) + I(t) + \eta P(t) - ETc(t) \quad (1)$$

where $\theta(t)$ is soil moisture (% volumetric water content) at step t, $I(t)$ is the irrigation dose applied by the actuation layer, $P(t)$ is rainfall, η is infiltration efficiency ($\eta = 0.7$ in this study), and $ETc(t)$ is crop evapotranspiration, the model's loss term.

Crop evapotranspiration follows the reference-evapotranspiration × crop-coefficient formulation established by the FAO-56 Penman–Monteith guideline [9], modified here so that the shading and additive treatments act on the loss term directly,

$$ETc(t) = ET0(t)(1 - Sr) / (1 + \beta\, Ar) \quad (2)$$

where $ET0(t)$ is reference evapotranspiration, $Sr \in [0, 1)$ is the fractional reduction in evaporative demand attributable to the shade treatment, $Ar \geq 0$ is the fractional increase in effective water-holding capacity attributable to the additive treatment, and β is a retention-coupling coefficient ($\beta = 0.55$) linking additive-driven water retention to a slower moisture-depletion rate. Setting $Sr = Ar = 0$ recovers the unshaded, unamended control condition.

B. Two-Layer Closed-Loop Control Law

The irrigation actuator state $v(t) \in \{0, 1\}$ is governed by a hysteresis threshold rule overridden by a rain-aware predictive layer and a manual-override input, evaluated in the order shown in Fig. 3,

$$v(t) = 0, \ \text{if override or } rain(t) = 1 \quad (3)$$

otherwise, with hysteresis between a dry threshold θdry and a wet threshold θwet ($\theta wet > \theta dry$) to prevent rapid on/off chattering at the boundary,

$$v(t) = 1 \text{ if } \theta(t) < \theta dry;\ 0 \text{ if } \theta(t) > \theta wet;\ v(t-1) \text{ else} \quad (4)$$

The rain-detection term $rain(t)$ is produced by a logistic classifier over the rain-sensor raw reading x (Section IV-C),

$$P(rain=1|x) = 1 / (1 + e^{-(\beta 0 + \beta 1 x)}) \quad (5)$$

with rain(t) = 1 when P(rain=1|x) exceeds a decision threshold of 0.5. Independently of the control law above, every actuation cycle is subject to a firmware-enforced safety constraint,

$$v(t) = 0 \text{ if } (t - t_{open} > T_{max}) \text{ or } (\Sigma T_{24h} > T_{day}) \quad (6)$$

where T_{max} bounds continuous valve-open time and T_{day} bounds cumulative daily watering time, so that neither a stale network command nor a control-logic fault can leave the valve open indefinitely a requirement does not present in passive-monitoring IoT designs [1], [2] but necessary once the system can act on its own data.

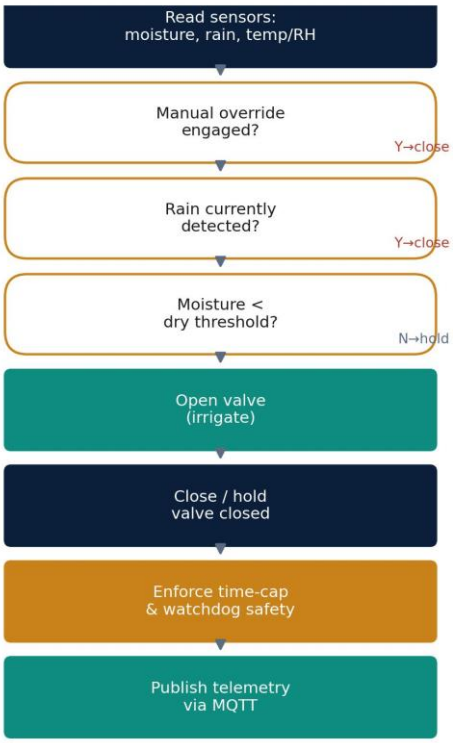


Fig. 3. Closed-loop irrigation decision workflow executed once per control cycle on the ESP32 node.

C. Factorial Statistical Model

The 3×3 factorial design (shade level i, additive level j, replicate k) is analyzed with a two-way fixed-effects ANOVA model,

$$y_{ijk} = \mu + \tau_i + \gamma_j + (\tau\gamma)_{ij} + \varepsilon_{ijk} \quad (7)$$

where y_{ijk} is the response (total irrigation water applied over the trial window), τ_i and γ_j are the shade and additive main effects, $(\tau\gamma)_{ij}$ is their interaction, and $\varepsilon_{ijk} \sim N(0, \sigma^2)$ [11]. A Random Forest regressor is additionally fitted over one-hot-encoded treatment levels to rank predictor importance by mean Gini-impurity reduction across its ensemble of trees [10], providing a non-parametric cross-check on which factor the ANOVA identifies as dominant.

D. Connectivity and Cloud Layer

Field nodes publish telemetry over Wi-Fi/MQTT for a single compact site, or through a LoRaWAN gateway when plots are distributed across a multi-acre area beyond reliable Wi-Fi range, consistent with reported LoRaWAN precision-agriculture deployments [5]. MQTT was selected as the application-layer protocol for it publish/subscribe model and low per-message overhead relative to synchronous alternatives, both properties established in prior lightweight-protocol evaluations for resource-constrained IoT devices [6]. A Mosquitto broker feeds a time-series store and dashboard, from which the factorial dataset in Section V is exported for analysis.

IV. EXPERIMENTAL SETUP AND METHODOLOGY

A. Hardware Platform

Each field node is built around a 30-pin ESP32 development module (Espressif Systems), selected over an Arduino Mega for its integrated Wi-Fi radio, avoiding a separate networking shield [8]. The node instruments one plot with a capacitive soil-moisture probe, a waterproof DS18B20 soil-temperature sensor, a DHT22 air temperature/humidity sensor, and a resistive rain sensor, and actuates a 12 V solenoid valve through a relay module switched from a digital GPIO, with the valve circuit electrically isolated from the ESP32 logic rail (Fig. 4).

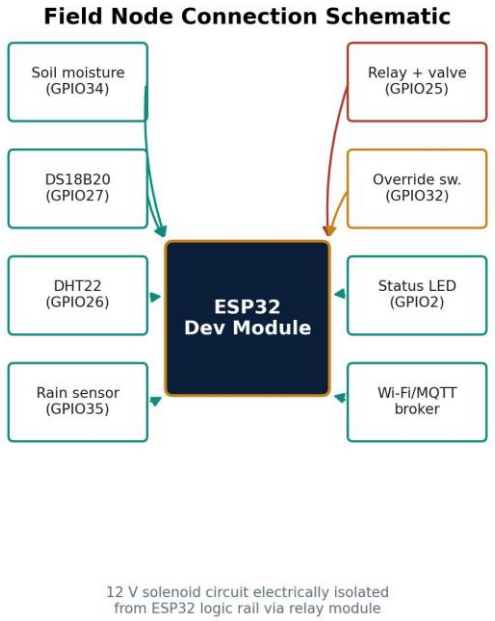


Fig. 4. Field-node hardware connection schematic; the 12 V actuation circuit is isolated from the ESP32 logic rail via the relay module.

B. Simulation Environment and Tools

The water-balance model of Section III-A and the control law of Section III-B were implemented in Python 3.11 (NumPy, SciPy) as the primary simulation environment. Statistical analysis used statsmodels for the two-way ANOVA (Section III-C) and scikit-learn for the Random Forest regressor and the logistic rain-detection classifier. This numerical-simulation approach was chosen over a circuit-level simulator (e.g., Proteus) because the object of validation here is the closed-loop control and factorial-treatment logic which depends on soil, weather, and treatment parameters rather than component-level electrical behavior while the embedded firmware itself (Section III-B logic, implemented in the accompanying Arduino/ESP-IDF sketch) was verified separately by tracing its decision logic against the same 48-hour reference scenario shown in Fig. 5, confirming that the firmware state machine reproduces the modeled control law.

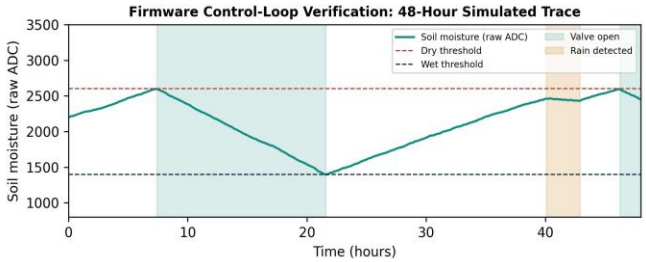


Fig. 5. 48-hour firmware control-loop verification trace: the controller irrigates once the dry threshold is crossed and correctly withholds a second irrigation cycle once rain is detected near 41 hours .

C. Factorial Design and Parameterization

The full factorial crosses three shade levels (bare control, biodegradable rice-husk mulch, and a plastic-mulch benchmark) with three additive levels (untreated control, a peel-derived hydrogel additive, and a synthetic superabsorbent-polymer benchmark), replicated three times, for 27 simulated plot-days series. Treatment parameters were set as literature-informed modeling assumptions rather than

fitted to any single reported value: the shade-driven evapotranspiration reduction Sr was set to 22% for the biodegradable mulch and 33% for the plastic-mulch benchmark, within the range of evaporation reductions generally reported for organic and plastic mulching [3]; the additive-driven water-holding-capacity increase Ar was set to 35% for the peel-gel additive and 42% for the synthetic-polymer benchmark, conservatively below the peak laboratory swelling ratios reported for peel-derived hydrogels [4] to remain plausible once diluted into a field soil mixture. Table I summarizes the simulation parameters.

Table I. Water-Balance Simulation Parameters

Parameter	Value / Range
Trial duration	30 days
Replicates per treatment	3 (27 series total)
Reference ET0 (baseline)	4.2 ± 0.4 mm/day
Field capacity (base loam)	34% VWC
Wilting point	12% VWC
Irrigation dry/wet threshold	20% / 30% VWC
Infiltration efficiency η	0.70
Retention-coupling coefficient β	0.55
Simulated rainfall events	5 events, 6–14 mm each
Max continuous valve-on (Tmax)	20 min
Max daily valve-on (Tday)	90 min

D. Rain-Classifier Training Setup

The rain-detection logistic classifier of Eq. (5) was trained on 4,000 simulated rain-sensor readings drawn from two overlapping Gaussian distributions (rain: $\mu = 1500$, $\sigma = 420$; no-rain: $\mu = 2550$, $\sigma = 430$ raw ADC units) representative of sensor noise and boundary ambiguity observed in low-cost resistive rain sensors, using a 75/25 train–test split evaluated by accuracy, precision, recall, and F1-score (Section V-D).

E. Testing Conditions

Two evaluation conditions were used throughout Section V: (i) a 48-hour single-node trace (Fig. 5) verifying firmware control-loop correctness at fine time resolution, and (ii) the full 30-day, 27-series factorial run (Section III-C) used for the statistical treatment-effect analysis. Both conditions share the same underlying water-balance and control-law equations (Section III), differing only in duration and in whether one plot or the full factorial matrix is simulated.

V. RESULTS AND SIMULATION

A. 30-Day Moisture Trends

Fig. 6 shows simulated 30-day moisture trajectories for four representative treatment combinations. The control×control series crosses the irrigation threshold most frequently, the bio-shade-only series shows a visibly slower decline between waterings, and the combined bio-shade×peel-gel series remains within the target band for the longest intervals of the four series shown, consistent with the two loss-reducing mechanisms (Sr and Ar in Eq. (2)) acting together rather than redundantly.

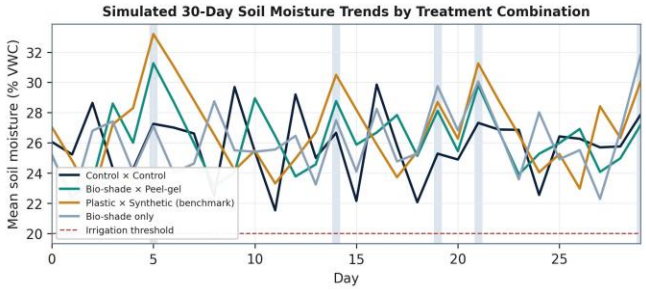


Fig. 6. Simulated 30-day mean soil-moisture trajectories for four representative shades additive combinations; shaded vertical bands mark simulated rainfall events.

B. Two-Way ANOVA on Irrigation Water Use

Table II reports the two-way ANOVA on total 30-day irrigation water applied per plot. Both main effects are statistically significant shade ($F(2,18) = 92.83$, $p < 0.001$) and additive ($F(2,18) = 29.66$, $p < 0.001$) while the shade×additive interaction is not significant ($F(4,18) = 0.72$, $p = 0.588$). This indicates that, within the parameter ranges simulated, the two treatments reduce irrigation demand through independent, additively combining mechanisms rather than a synergistic or redundant interaction, directly answering the question neither single factor reviewed in Section II can address.

Table II. Two-Way ANOVA on Total Irrigation Water Applied (mm)

Source	df	F	p-value
Shade	2, 18	92.83	< 0.001
Additive	2, 18	29.66	< 0.001
Shade × Additive	4, 18	0.72	0.588

C. Random Forest Feature Importance

The Random Forest regressor (Section III-C) attributed 75.8% of its aggregated Gini-importance to the shade factor and 24.2% to the additive factor in predicting total irrigation water use, corroborating the ANOVA finding that shade is the dominant single driver of water demand in this simulation, with the additive contributing a smaller but statistically confirmed independent reduction.

D. Water-Savings Comparison

Table III and Fig. 7 summarize simulated irrigation water savings relative to the unshaded, unamended control. The fully biodegradable combination (bio-shade × peel-gel) achieved 45.0% simulated water savings, compared with 57.8% for the fully synthetic benchmark combination (plastic mulch × synthetic polymer) that is, the biodegradable combination recovered 77.9% of the benchmark's water-saving performance while avoiding persistent plastic and synthetic-polymer field residues [3]. The single best-performing combination overall, plastic mulch × peel-gel, reached 58.3% savings, indicating that the additive's benefit is not conditional on a particular shade material, consistent with the non-significant interaction term in Table II.

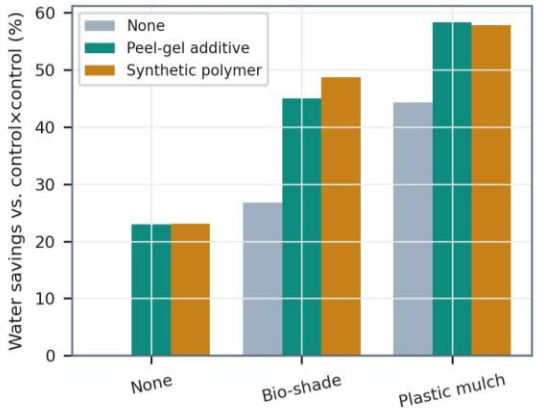


Fig. 7. Simulated irrigation water savings (%) relative to the control×control baseline, grouped by shade treatment and colored by additive treatment.

Table III. Simulated 30-Day Irrigation Water Use and Savings by Treatment Combination

Shade × Additive	Water use (mm)	Savings (%)
Control × Control	89.1 ± 5.7	0.0
Control × Peel-gel	68.7 ± 2.5	22.9
Bio-shade × Control	65.2 ± 6.3	26.8
Bio-shade × Peel-gel	49.0 ± 6.5	45.0
Plastic × Control	49.6 ± 7.4	44.3
Bio-shade × Synthetic	45.7 ± 1.2	48.7
Plastic × Synthetic (benchmark)	37.6 ± 5.6	57.8
Plastic × Peel-gel (best overall)	37.2 ± 7.2	58.3

E. Interaction Plot

Fig. 8 plots mean irrigation water use across shade levels for each additive level; the three near-parallel lines visually corroborate the ANOVA's non-significant interaction term (Table II) the additive shifts each curve downward by a broadly similar margin regardless of which shade treatment it is paired with.

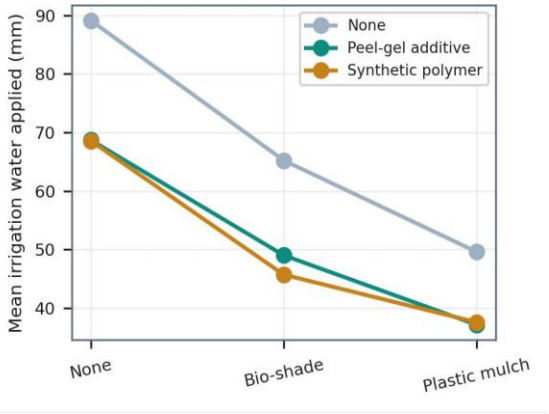


Fig. 8. Interaction plot of mean 30-day irrigation water use by shade level, grouped by additive; near-parallel lines indicate the absence of a significant interaction effect.

F. Rain-Detection Classifier Performance

The logistic rain-detection classifier (Section IV-D) achieved 87.4% accuracy, 88.6% precision, 86.5% recall, and an F1-score of 0.876 on 1,000 held-out test samples (Fig. 9). The residual error rate reflects the deliberately overlapping sensor-noise distributions used in training (Section IV-D) and represents the predictive layer's expected reliability under realistic low-cost sensor noise rather than an idealized, perfectly separable input.

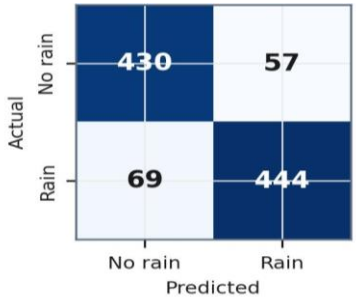


Fig. 9. Confusion matrix for the rain-detection logistic classifier on the held-out test set.

VI. CONCLUSION

This paper presented AquaShield-AI, a closed-loop IoT framework that jointly evaluates biodegradable shading and soil-additive treatments within a 3×3 factorial design and actuates irrigation through a two-layer, fail-safe ESP32-based controller. Simulation results, analyzed by two-way ANOVA, showed statistically significant independent main effects of shade ($p < 0.001$) and additive ($p < 0.001$) on irrigation water use with no significant interaction ($p = 0.588$), indicating that the two biodegradable interventions combine additively rather than redundantly. The fully biodegradable treatment combination achieved 45.0% simulated water savings 77.9% of the 57.8% achieved by a fully synthetic benchmark while a rain-aware predictive control layer reached 87.4% classification accuracy in withholding unnecessary irrigation ahead of rainfall. Random Forest feature importance corroborated the ANOVA, attributing three-quarters of the explained variance in water use to the shade factor and one-quarter to the additive factor.

The principal limitation of this work is that the reported results are simulation-based: treatment-effect magnitudes were set as literature-informed modeling assumptions rather than fitted to measurements from a physical field trial, and the firmware control loop was verified against a modeled reference trace rather than deployed hardware operating under real weather and soil variability. The factorial sample size ($n = 3$ per cell) is also modest for detecting a smaller true interaction effect than the one simulated here, so the reported absence of interaction should be treated as a simulation-supported hypothesis for field verification, not a settled empirical conclusion.

Future work should prioritize three directions: physical deployment of the full 3×3 factorial across an instrumented field site to replace the simulated treatment-effect parameters with measured values; extension of the connectivity layer to a multi-node LoRaWAN

deployment to test the architecture at farm scale beyond the single-gateway range assumed in Section III-D; and an economic viability analysis comparing the fully instrumented research-tier system evaluated here against a lower-cost, non-connected adoption-tier variant, to determine which configuration is realistically deployable by the smallholder farmers this system is ultimately intended to serve.

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